

UNIT III - Agricultural Systems Management

1. Define an agricultural system.

An agricultural system is a group of components, functions and processes working together to achieve a clear purpose. Example: a crop production system.

2. Mention the role of an Agricultural Systems Manager.

An Agricultural Systems Manager plans, evaluates and adjusts the agricultural system and its components to achieve the required goals.

3. List any four characteristics of an agricultural system.

- A clear purpose
- Defined objectives
- Measurable performance and
- Connected components and processes are important characteristics.

4. Give an example of a well-defined agricultural system.

A potato-chip production system is an example. It may have targets for production rate, waste, quality, safety and cost.

5. Define reliability in an agricultural system.

Reliability is the probability that a system performs successfully under specified conditions. It is usually expressed as a value between 0 and 1.

6. Mention two ways to improve system reliability.

1. Reliability can be improved by using reliable components
2. Providing backup or redundant units.

7. What is redundancy in an agricultural system?

Redundancy means keeping an additional component or unit as a backup so that the system can continue when one unit fails. Example: keeping a backup tractor.

8. Differentiate serial and parallel operation of components.

Serial	Parallel
All required units must work	One of the available units can perform the function
Failure of one required unit can stop the system	Backup can improve reliability

9.State the use of simulation in agriculture.

Simulation is used to represent agricultural processes on a computer and study their behaviour. It can be used for crop growth and field-operation planning.

10. Mention two uses of crop-growth simulation.

It can help study crop growth and predict the effect of different farming decisions on crop performance.

11. How does field-operation simulation help farmers?

It helps study operations such as planting, harvesting and machinery use before carrying them out, helping managers compare possible plans.

12. Define resource optimization.

Resource optimization means using available resources such as land, labour, water, machinery and money in the best possible way.

13. What is linear programming?

Linear programming is a mathematical technique used to find the best solution when resources are limited and objectives and constraints can be expressed using linear relationships.

14. Define project scheduling.

Project scheduling is the planning and arrangement of project activities according to their order, time and available resources so that the project can be completed on time.

15. Mention the main uses of AI and Decision Support Systems in agriculture.

AI can help with prediction and analysis. Decision Support Systems help with diagnostics, marketing, operations management, automatic control and strategic planning.

UNIT IV - Weather Prediction Models

1. Define climate variability.

Climate variability means changes in climate conditions such as temperature and rainfall over different time periods.

2. Mention the two main types of climate variability.

1. Natural variability
2. Human-induced (anthropogenic) variability.

3. Give two examples of natural climate variability.

- El Niño and La Niña
- Volcanic eruptions
- Solar cycles are examples of natural climate variability.

4. Mention two human activities that affect climate variability.

Burning fossil fuels and deforestation can change greenhouse gas levels and land-use patterns.

5. Define seasonal forecasting.

Seasonal forecasting is the prediction of climate conditions such as temperature and rainfall for the coming months, usually from one month to one year.

6. Mention two uses of seasonal forecasting in agriculture.

It helps farmers plan planting and harvesting and prepare irrigation or other measures based on expected climate conditions.

7. State two uses of seasonal forecasting in water management.

- It helps plan water supply and manage reservoirs
- It also helps prepare for possible droughts and floods.

8. Mention two uses of seasonal forecasting for disaster management.

- It provides early information about possible extreme weather.
- It helps organizations prepare response plans and improve infrastructure.

9. Define Global Climatic Model (GCM).

A GCM is a computer-based model that represents the physical processes and interactions of the atmosphere, oceans, land surface and ice.

10. List the main components of a GCM.

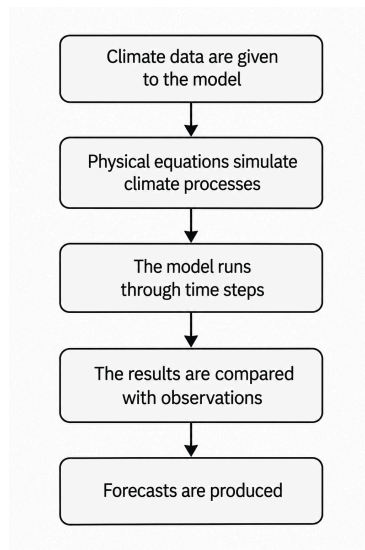
The main components are

- atmospheric models
- ocean models
- land-surface models
- cryospheric models.

11. What is the grid system in a GCM?

A GCM divides the Earth into grid cells. Each cell represents a geographical area, and the cell size determines the model resolution.

12. Outline the basic working of a GCM.



13. Mention any three potentials of GCMs for seasonal forecasting.

GCMs can provide dynamic predictions, help predict El Niño and La Niña, give regional information through downscaling, and support multi-model ensemble forecasts.

14. List two limitations of GCMs.

- GCMs have uncertainty and coarse resolution for some local features.
- High-resolution models also need large computing resources.

15. Define the General Systems Approach for applying seasonal forecasts.

It is a framework that views climate, agriculture, water, social and economic factors as connected parts of one system. It uses their interactions to support better decisions.

UNIT V - E-Governance in Agricultural Systems

1. Define an expert system.

An expert system is a computer program that uses stored expert knowledge and rules to solve problems and support decisions in a specific field.

2. Mention the main characteristics of an expert system.

- High performance
- Understandable responses
- Reliability
- Quick responses

3. List the three main components of an expert system.

1. User interface
2. Inference engine
3. Knowledge base.

4. What is an inference engine?

The inference engine is the main processing part of an expert system. It applies rules to the knowledge base to derive a conclusion.

5. Differentiate forward chaining and backward chaining.

Forward chaining starts with known facts and applies rules to reach a conclusion. Backward chaining starts with a goal and works backward to find supporting facts.

6. Define a knowledge base.

A knowledge base stores the facts, rules and expert knowledge needed by an expert system to solve problems.

7. Differentiate factual knowledge and heuristic knowledge.

Factual knowledge	Heuristic knowledge
Based on accepted facts	Based on experience and practice
More direct and known	Uses practical judgement or rules

8. Define a Decision Support System (DSS).

A DSS is a computer-based system that uses data, models or knowledge to help a person make an informed decision.

9. Mention the main types of Decision Support Systems.

- Document-driven DSS
- Data-driven DSS
- Knowledge-driven DSS
- Model-driven DSS
- Communication-driven DSS.

10. What are agricultural and biological databases?

They are digital repositories that store agricultural and biological information such as crops, livestock, genetics, soil, weather, pests and diseases.

11. Mention four uses of agricultural and biological databases.

They support,

- Data collection
- Crop and livestock information
- Soil and weather analysis
- Pest and disease management
- Research and decision support.

12. Define e-commerce in agriculture.

E-commerce means buying and selling goods and services through the internet. In agriculture, it connects farmers, buyers and consumers through digital platforms.

13. Mention four applications of e-commerce in agriculture.

- Online marketplaces
- Purchase of agricultural inputs
- Direct sales to consumers
- Digital payments
- Supply-chain management
- Market information
- Online advisory services

14. Define e-business in agricultural systems.

E-business means using digital technologies for business activities such as buying, selling, marketing and managing operations in the agricultural value chain.

15. Differentiate e-commerce and e-business.

E-commerce	E-business
Mainly online buying and selling	Covers wider digital business activities
Part of e-business	Includes e-commerce and other operations

